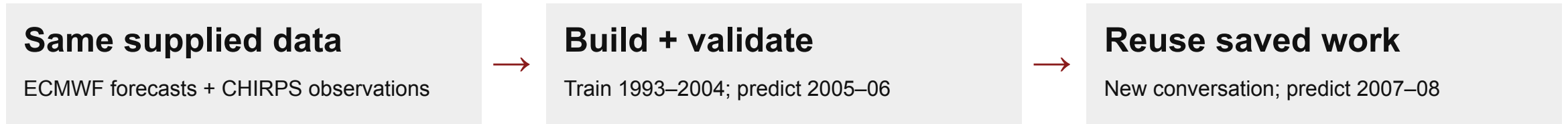


Can coding agents build a reusable seasonal forecast?

Task: turn September-issued forecasts into calibrated October–December rainfall probabilities for 12 grid cells in Kenya, then reuse the saved workflow for the next forecast batch.



Three arms: scratch, Rhiza skills, ACCORD (DeepScale / Rosetta)

1 Agent capability

Can a stronger model complete the workflow from scratch?

2 Skills and libraries

Does Rhiza or ACCORD make the same task faster and cheaper?

3 Maintained code

Does saved code and fitted state reduce work on the next forecast?

Evaluate completion, scientific quality and effort separately

12 artifact / calculation / replay checks + evidence-based LLM review; forecast error (RPS, RMSE);

agent tokens, elapsed time and API cost. Same inputs and limits; only tools and model vary.

What changes with model capability and tools?

Seven initial attempts on the same task. Read completion alongside tokens, time and cost.

Model / tools	Checks passed / 12	Agent API cost (\$)	Tokens	Minutes
Haiku / scratch	3/12	\$0.92	776,678	4.5
Haiku / Rhiza	7/12	\$0.86	748,020	3.8
Sonnet / scratch	9/12	\$0.26	86,410	1.6
Sonnet / Rhiza	12/12	\$0.36	132,946	1.6
Fable / scratch	12/12	\$3.36	260,515	4.9
Fable / Rhiza	11/12	\$4.42	360,047	4.9
Fable / ACCORD	12/12	\$4.39	365,355	4.1

Sonnet + Rhiza matched Fable scratch’s 12/12 checks: 49% fewer tokens, 67% less time and 89% lower API cost. Fable + ACCORD was 17% faster than scratch.

Cost bars share a \$0–\$5 scale. One attempt per condition. Sonnet + ACCORD: not run.

Sonnet scratch missed unit labels. Rhiza supplied catalog context, without script calls. Valid forecasts had similar initial RPS ($\approx 0.279\text{--}0.281$).
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Does saved work make the next forecast cheaper?

All three Fable arms keep their code, cached inputs and fitted model; the next conversation is fresh.

Agent API cost (\$) · common scale \$0–\$5

■ Initial

■ Follow-up

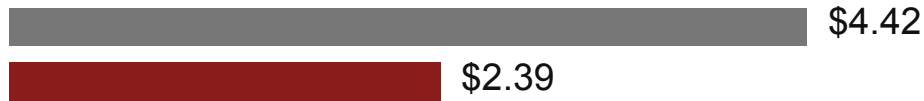
Scratch



Follow-up: tokens / time / checks

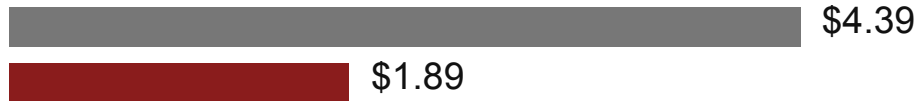
334,053 / 3.8 min / 12/12

Rhiza



213,139 / 2.5 min / 11/12

ACCORD



165,576 / 2.6 min / 12/12

ACCORD's follow-up used 50% fewer tokens, cost 50% less and took 31% less time than scratch; both passed 12/12 checks. Across both attempts, ACCORD cost 12% less.

Follow-up RPS ↓: scratch 0.396; Rhiza 0.404; ACCORD 0.412. No accuracy advantage established.

All arms reused fitted models. No fresh-workspace controls; no download savings measured; two development years per batch.

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